

QoE-Aware Deep Reinforcement Learning for URLLC-Prioritized Slice Admission Control: From Offline Simulation to a Live O-RAN Testbed

Manoj Prasad Kunasegran, Wai Leong Pang, and Swee King Phang

TODO(MANOJ): author order/initials carries the same authorship before

School of Engineering, Faculty of Innovation & Technology, Taylor's University, Subang Jaya, 47500, Selangor, Malaysia **TODO(MANOJ)**

Abstract—Deep reinforcement learning (DRL) is widely proposed for O-RAN slice admission control, but most of this work – including our own [1], [2] – is validated only in simulation, a gap our systematic review of 67 core 2024–2026 studies formalized as the field’s *validated-deployment gap* [3]. This paper narrows that gap – realizing the control loop on real protocol-stack, MAC-scheduler, and E2-agent hardware, radio channel still simulated (rfsim) – and centers the evaluation on quality of experience (QoE): a QoE-aware DQN admission policy ($r = \alpha \cdot \text{MOS} - \beta \cdot \text{cost} - \xi \cdot \text{SLA_viol}$) deployed on a real OpenAirInterface (OAI) gNB across three slices, against a static baseline, its SLA-only counterpart, and a static-at-cap oracle that removes the admission decision itself (every ceiling parked at its calibrated maximum, never adapted). Live, DQN holds $100.0 \pm 0.0\%$ SLA compliance against the baseline’s seed-dependent $73.6 \pm 18.6\%$ (worst episode 0.0%) across 45 episodes, and – unlike the oracle, which still collapses into the same catastrophic-backlog regime in 4 of 15 episodes ($p = 0.0149$, DQN 25/25 vs. oracle 11/15 fully compliant over a reverification round) – this rules out “DQN just rediscovers a fixed high ceiling” as an explanation for the result. We disclose the engineering numbers a real deployment needs (sub-millisecond E2 round trip, sub-100- μs inference, a control loop holding its configured 5 s cadence to within 1 ms), and report two independent MOS-calibration bugs found and fixed mid-study (a throughput units mismatch; QoE scored against generic rather than each slice’s configured SLA deadline), with a directional, not yet fully-powered, live re-run under the correction. A held-out, offline-only experiment under genuinely congested, dynamic, randomized multi-slice traffic with real shared-PRB contention shows DQN, once trained with a correctly decaying exploration schedule, preserves URLLC SLA compliance under either reward ($22.6\% \rightarrow 30.9\%$ SLA-only, $\rightarrow 27.0\%$ QoE-aware) by trading off eMBB ($\rightarrow 7.9\%/8.2\%$) and mMTC ($\rightarrow 9.2\%/10.8\%$) – the URLLC-first outcome our prior simulated work targeted, now shown under real cross-slice scarcity and a property of the trained policy rather than the specific reward formulation, though not yet a win on this paper’s own priority-weighted utility metric. We report where offline-to-live transfer does and does not hold, and position this evidence against the five validation directions our review’s proposed QoE-aware GAT-MARL-FL framework calls out.

Index Terms—O-RAN, network slicing, deep reinforcement learning, admission control, quality of experience, URLLC prioritization, live testbed

I. INTRODUCTION

Network slicing lets one physical O-RAN deployment serve heterogeneous service classes (eMBB, URLLC, mMTC)

under per-slice SLA guarantees, but static, threshold-based control cannot adapt PRB allocation as demand shifts [4]. DRL is a natural fit: our own prior work formulated single-gNB SLA-aware slice admission control [1] and extended it to a third slice with a multi-gNB load-balancing term [2], reporting $\approx 99.5\%$ and $99.63/88.41/97.64\%$ per-slice URLLC/eMBB/mMTC SLA satisfaction respectively – both validated entirely in simulation. Our systematic review of the 2024–2026 literature [3] (6,286 records screened to 67 core studies) found this pattern holds across the field and crystallized it as one of five recurring gaps – alongside QoE inference, topology awareness, privacy preservation, and scalability – motivating a proposed QoE-aware GAT-MARL-FL framework (graph-attention state encoding [5], centralized-training multi-agent control, and federated learning [6] across gNBs) to close them.

This paper makes five contributions. (1) We realize the slice admission-control MDP from [1], [2] on a real OAI gNB over its native, RIC-free E2 control loop – not simulation – directly addressing the validated-deployment gap, and surface a structural finding for anyone attempting this on real O-RAN hardware: the real E2 agent exposes no per-flow accept/reject primitive, only a per-slice PRB ratio ceiling, so admission must be realized as *ceiling nudging*. (2) We center the evaluation on QoE, comparing a QoE-aware reward directly against the SLA-only reward of [1], [2] on the same live hardware. (3) We report offline-simulated and live results side by side for the same frozen policies, showing where simulation-to-real transfer holds and where it does not. (4) We add a config-only *static-at-cap* oracle arm – every ceiling parked at its calibrated maximum, never adapted – to isolate genuine admission timing from just riding every ceiling upward, and disclose the engineering numbers and two QoE/MOS calibration bugs a real deployment surfaces that offline work does not. (5) We build a genuinely congested, dynamic, randomized multi-slice scenario with real shared-PRB contention and show a properly-trained DQN policy, under either reward, learns to preserve URLLC at the deliberate expense of eMBB/mMTC – the same URLLC-first objective as [1], [2], now under real cross-slice scarcity rather than isolated per-slice utility.

Scope. Single gNB, three UEs (one per slice), rfsim. The

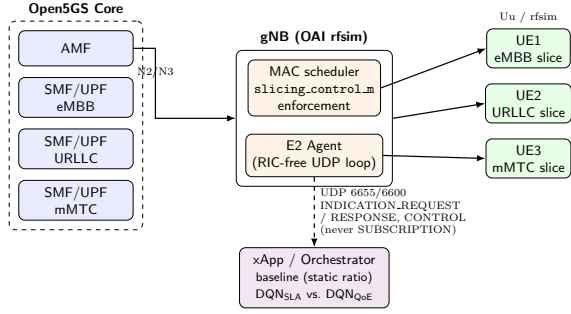


Fig. 1: Testbed control loop: Open5GS core, OAI gNB (MAC scheduler + E2 agent), 3 rfsim UEs (one per slice), and the xApp/orchestrator realizing baseline vs. DQN.

three slices already contend for one shared 106-PRB pool – a genuine, small-scale multi-agent coordination problem. An explicit multi-agent (GAT-MARL) controller over that pool, multi-gNB coordination, and federated learning – our review’s proposed framework in full [3] – are the journal-length follow-up’s target (Section V), not attempted here.

II. SYSTEM ARCHITECTURE AND TESTBED

Fig. 1 shows the control loop: an Open5GS core with three slice-specific SMF/UPF pairs, an OAI gNB (ORANSlice fork [7], RIC-free UDP E2 agent built into the gNB binary, distinct from vanilla OAI’s FlexRIC [8], [9]), and three nr-uesoftmodem UEs, one per slice. The E2 agent’s only control primitive is `slicing_control_m`{sst, sd, min_ratio, max_ratio} [10], [11] – a per-slice PRB ratio ceiling enforced by the MAC scheduler, with no raw per-flow accept/reject hook. The binary admission decision of [1], [2] is therefore realized as ceiling nudging: accept raises a slice’s `max_ratio` toward a calibrated cap, reject lowers it toward its floor. Ratio caps were calibrated by direct live measurement, not assumed linear (Table I): the 0–100 ratio scale does not map linearly onto real PRBs at this cell’s 106-PRB/ $\mu=1$ configuration, so each slice’s cap was set empirically below its measured organic demand. The static baseline pins each ceiling-step size to zero (no accept/reject can move it all episode); learned arms are DQN [12] policies trained offline under an SLA-only (Eq. 1) or QoE-aware reward (Eq. 2) against a closed-loop synthetic KPM source, then evaluated live with frozen weights.

III. PROBLEM FORMULATION

Each admission decision at step t accepts or rejects incoming demand for slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, realized as the ceiling adjustment above. Two reward formulations are compared. The SLA-only reward (eq. 2 of [1], [2]):

$$r_t^{\text{SLA}} = \sum_{s \in \mathcal{S}} w_s c_s n_{s,t}^{\text{acc}} - \sum_{s \in \mathcal{S}} \mathbb{I}[v_{s,t}] p_s - \eta \bar{\rho}_t N_t^{\text{acc}} \quad (1)$$

where w_s/p_s are slice s ’s priority weight and violation penalty (URLLC highest of the three, Table I), $n_{s,t}^{\text{acc}}$ is slice s ’s accepted count, $\mathbb{I}[v_{s,t}]$ flags an SLA violation, and $\eta \bar{\rho}_t N_t^{\text{acc}}$ penalizes total accepted volume by mean cluster congestion $\bar{\rho}_t$. The

TABLE I: Testbed and campaign parameters

Parameter	Value
gNB / E2 primitive	OAI rfsim, 106 PRB, $\mu=1$; <code>slicing_control_m</code> ratio ceiling
Slices	eMBB (4 Mbps), URLLC (300 kbps), mMTC (50 kbps bursty, 2 s/6 s)
Ceilings (eMBB/URLLC/mMTC)	12 / 4 / 3 (below each slice’s measured organic demand)
QoE weights α, β, ξ	1.0, 0.2, 0.5 (Eq. 2)
Arms	baseline (static), DQN _{SLA} , DQN _{QoE}
Live episodes	3 seeds $\times$ 5 ep./arm = 45 learned + 15 baseline
Offline training	300 ep. $\times$ 3 seeds/arm; frozen weights at live deployment

QoE-aware reward (eq. 3 of [3]), used to center this paper’s evaluation on QoE rather than SLA alone:

$$r_t^{\text{QoE}} = \alpha \widehat{\text{MOS}}_t - \beta \bar{\rho}_t N_t^{\text{acc}} - \xi \overline{\text{SLA_viol}}_t \quad (2)$$

with $\alpha=1.0, \beta=0.2, \xi=0.5$ (matching [3]’s own symbol for the SLA-violation weight) and $\widehat{\text{MOS}}_t \in [0, 1]$ the mean inferred per-slice MOS, normalized from its native P.1203/ACR 1–5 scale via $(\text{MOS}-1)/4$ (Table II reports the raw, un-normalized 1–5 values). MOS itself is estimated with a Weber–Fechner-law closed-form prior – eq. 4 of [3], $\text{MOS} = a - b \ln(1 + Q_{\text{deg}})$, distinct from the exponential IQX-hypothesis QoE-mapping family [13] – refined by a per-slice LSTM, calibrated against ITU-T P.1203 [14] objective labels for eMBB and ACR-style task-success scoring for URLLC/mMTC, and $\overline{\text{SLA_viol}}_t \in [0, 1]$ the mean per-slice violation *severity* (continuous margin, not a flat per-violation charge). Section IV-E’s congested scenario additionally enforces a genuine shared-PRB-pool constraint absent from the base offline simulator: each slice’s requested PRBs $\text{req}_{s,t}$ are served in full only while the pool has slack, otherwise every slice is rationed proportionally to its own request share,

$$\text{served}_{s,t} = \begin{cases} \text{req}_{s,t}, & \sum_{s'} \text{req}_{s',t} \leq B \\ \text{req}_{s,t} \cdot \frac{B}{\sum_{s'} \text{req}_{s',t}}, & \text{otherwise} \end{cases} \quad (3)$$

with B the shared PRB budget. This rationing rule is priority-neutral by construction: any URLLC protection observed downstream must come from the policy’s reward-driven behavior, not from the physics.

Raw per-slice compliance treats every slice as equally important, which Table I’s own w_s (priority weight) says is not the case. We therefore also report a priority-weighted SLA utility,

$$U = \frac{\sum_{s \in \mathcal{S}} w_s \cdot \text{compliance}_s}{\sum_{s \in \mathcal{S}} w_s}, \quad (4)$$

using each experiment’s own declared w_s (Table I), alongside the raw per-slice figures, not instead of them – U answers “how well does this arm do against the objective the reward actually declares,” which an unweighted mean across slices of unequal declared importance does not.

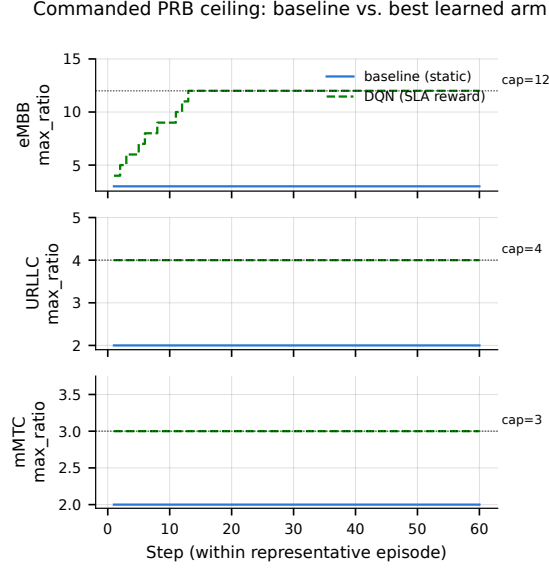


Fig. 2: Commanded PRB ceiling per slice over one representative episode: baseline (flat) vs. DQN (rides to the calibrated cap, annotated per slice).

IV. RESULTS

A. Live Testbed: DQN vs. Static Baseline

Fig. 2 shows the mechanism directly: baseline’s commanded ceiling is bit-for-bit flat at its static nominal ratio for the entire episode, while DQN rides eMBB’s ceiling from floor to its calibrated cap within ~ 13 steps and jumps URLLC/mMTC to their caps immediately, holding both for the remainder. The direct consequence, across 45 live episodes (3 seeds $\times$ 5 episodes $\times$ 2 reward modes) on real hardware: DQN under both the SLA-only and QoE-aware reward holds $100.0 \pm 0.0\%$ SLA compliance on every slice, in every episode, against the static baseline’s $73.6 \pm 18.6\%$ (bootstrap 95% CI [53.4, 93.4]), with a worst single episode of 0.0% for baseline versus 100.0% for DQN (Fig. 3, Table II).

Baseline’s compliance is real but *seed-dependent* (60.3%/100.0%/60.7% across the three seeds) – the practical value demonstrated is reliability under variable conditions, not rescuing an otherwise-always-failing baseline. We do not report Cohen’s d : every DQN arm’s per-seed compliance has zero variance, so a pooled-variance effect size would be driven entirely by baseline’s own spread. We omit per-slice inferred MOS from Table II: it was computed under a calibration later found, diagnosed, and fixed to carry a units bug specifically affecting eMBB/URLLC (Section IV-D reports the corrected picture and why the SLA-compliance numbers here are unaffected by it).

A natural objection: does DQN’s advantage reduce to simply parking every ceiling at a generous fixed value, something a differently-configured static policy could match without learning anything? We add *static-at-cap*, a config-only fourth arm that pins every ceiling at its calibrated cap from step zero (baseline’s own `AlwaysAcceptPolicy` entrypoint, unmodified) – the state DQN itself converges to and holds

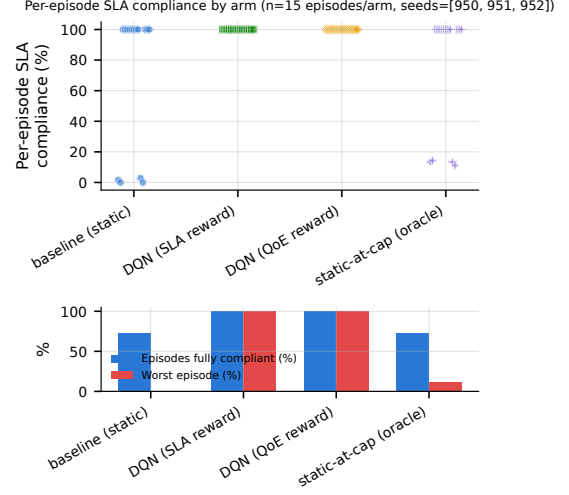


Fig. 3: Per-episode SLA compliance (%), $n = 15$ episodes/arm. Top: one point per episode – both DQN arms sit at exactly 100.0% in all 15 episodes; baseline and the static-at-cap oracle each show the same bimodal split. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance.

TABLE II: Live-campaign results, mean $\pm$ std over 3 seeds (worst ep. pooled across all episodes, not a mean-of-means). U = priority-weighted SLA utility, Eq. 4. static-at-cap: config-only oracle, no learning, no admission decision.

Arm	Slice	SLA compl. (%)	Worst ep. (%)	U (%)
baseline	eMBB	73.7 ± 18.6	0.0	73.5
	URLLC	73.4 ± 18.8	0.0	
	mMTC	73.8 ± 18.5	0.0	
DQN (SLA)	eMBB	100.0 ± 0.0	100.0	100.0
	URLLC	100.0 ± 0.0	100.0	
	mMTC	100.0 ± 0.0	100.0	
DQN (QoE)	eMBB	100.0 ± 0.0	100.0	100.0
	URLLC	100.0 ± 0.0	100.0	
	mMTC	100.0 ± 0.0	100.0	
static-at-cap	eMBB	83.1 ± 12.0	33.3	77.3
	URLLC	73.4 ± 18.8	0.0	
	mMTC	73.9 ± 18.5	0.0	

by episode end (Fig. 2). Fig. 3 and Table II show it is not equivalent to DQN: static-at-cap collapses into the same catastrophic backlog-blowup regime baseline exhibits in 4 of its 15 episodes (27%, statistically indistinguishable from baseline’s own 4/15 – Fisher exact $p = 1.0$), while DQN, re-evaluated across a reverification round on 5 distinct seeds (2 never used in the original campaign) for 25 episodes total, never collapsed once (static-at-cap 11/15 vs. DQN 25/25 fully compliant, Fisher exact $p = 0.0149$). In episodes where it does not collapse, static-at-cap’s severity distribution is statistically indistinguishable from DQN’s – **the mechanism is the admission decision, not the ceiling value**: neither baseline nor static-at-cap ever rejects under backlog pressure; DQN’s contribution is learning *when* to reject, which no fixed ceiling, however generously set, can do.

B. Engineering Instrumentation: Is This Deployable?

Measured directly against the real gNB E2 agent and the real, frozen checkpoints on CPU (no GPU on this near-RT path): E2 `poll()` round trip 0.566 ms median (1.773 ms p99); `send_control()` 0.168 ms (send-call overhead only – this E2 agent returns no response to CONTROL messages, fire-and-forget); DQN inference $68\mu\text{s}$ median ($115\mu\text{s}$ p99) on a 307 KB, 18,562-parameter checkpoint; control-loop cadence 5.0007 s measured against a configured 5.000 s – a naive estimate from batched subprocess wall-clock time had instead suggested a spurious $\approx 85\text{--}90\text{ ms/step}$ gap, traced to a fixed $\approx 8\text{--}9\text{ s}$ per-launch startup cost amortized over a batch’s episode count, not a per-step cost, and absent from every compliance/utility number in this paper. The per-step cost that does dominate is QoE inference, not RL decision-making: the MOS-mapper LSTM alone costs $\approx 7.3\text{ ms}$ against the DQN network’s $\approx 0.07\text{ ms}$ and the E2 protocol’s $\approx 0.6\text{ ms}$ – all under 2% of budget.

C. Offline vs. Online: Sim-to-Real Transfer

DQN is trained offline against a closed-loop synthetic KPM source, then evaluated live with frozen weights; Fig. 4 compares each arm’s offline-predicted compliance (final 5 rollup episodes/seed, pooled across 3 training seeds) against its live-measured compliance. The result is not a clean diagonal. eMBB transfers exactly: 100.0% $\rightarrow$ 100.0% for both reward modes. URLLC and mMTC do not: offline-predicted compliance is exactly 0.0% for both slices, both arms, throughout all 300 training episodes, while the same frozen policies measure 100.0% live – not an accidental scale mismatch: the offline training config deliberately pins URLLC/mMTC’s ceiling at nominal ratio with zero elastic headroom, an oversubscription-by-construction regime that never intended offline compliance to predict live behavior for these two slices. The SLA-only reward therefore had no compliance-based learning signal to differentiate policies on URLLC/mMTC offline – whatever separates live URLLC/mMTC behavior across arms was not something the offline reward could have selected for. This ceiling-headroom mismatch is distinct from, and not resolved by, Section IV-D’s later demand-scale fix; we flag it as open (Section VI), not as a result to spin.

D. QoE/MOS Recalibration: Two Bugs Found and Fixed

Section IV-A’s omitted MOS column reflects two calibration bugs found and fixed, not floor-pinned QoE. **Bug 1:** the frozen environment feeds `prb_used_ratio` (usage as a fraction of the gNB’s 100-PRB capacity) into the MOS mapper’s throughput input, but the frozen calibration script fit that input as an absolute PRB/UE count in $[0.05, 5.0]$ – eMBB’s real $\approx 15\text{-PRB}$ demand (ratio ≈ 0.15) read as almost-starved, not well-served near its 12-PRB ceiling. **Bug 2:** URLLC/mMTC’s MOS labels were scored against generic default SLA deadlines (5 ms/0.1%, 1000 ms/5%), not this campaign’s real thresholds (20 ms/0.5%, 65 ms/3.5%, Table I). Both fixed and verified at real operating points: eMBB MOS at ratio 0.15 moves $1.000 \rightarrow 3.624$ ($r=0.987$); URLLC MOS at its real 20 ms

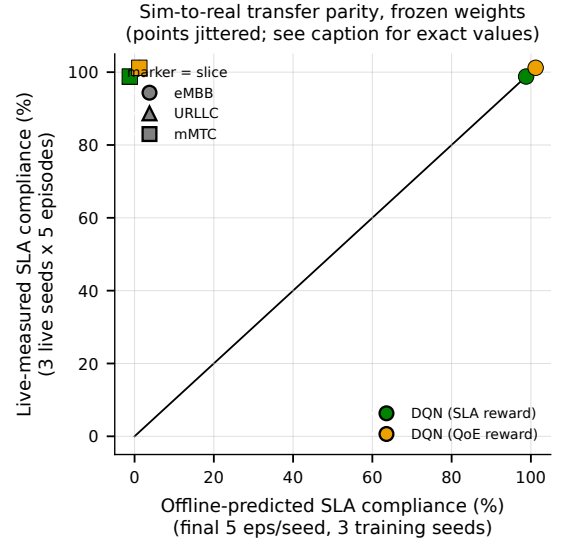


Fig. 4: Offline-predicted vs. live-measured SLA compliance, frozen weights, DQN under both reward modes (points jittered for visibility; both arms coincide exactly per slice). eMBB sits on $y=x$; URLLC/mMTC sit at (offline= 0, live= 100).

deadline moves $1.000 \rightarrow 3.525$. Neither bug touches the SLA-compliance/margin path Section IV-A’s numbers come from – only the omitted MOS reading needed correction.

A third, related bug surfaced while retraining under the fix: the offline environment’s demand-scale formula produces $\approx 4\times$ less mean load than real live traffic, leaving the corrected calibration nothing to differentiate. We fixed this with a new environment factory driven by real measured demand ratios, validated zero-training before spending compute, and retrained both DQN checkpoints under all three fixes. A directional live check (3 seeds $\times$ 2 episodes, cut by explicit time budget) reproduces Section IV-A’s qualitative picture (baseline $U=67.0\%$ vs. 100.0% for both DQN arms), MOS no longer floor-pinned (eMBB 1.91–2.17 vs. ≈ 1.2 before). A further round asked whether static-at-cap’s collapse-avoidance finding survives the correction: it replicates directionally (DQN 16/16 vs. static-at-cap 19/21) but not at significance ($p=0.495$ vs. $p=0.0149$ above) – reported as such, not rounded up. A fully-powered live re-run under this correction is this paper’s most concrete next experiment (Section VI).

E. Congested, URLLC-Prioritized Admission

Sections IV-A–IV-C use one constant per-slice demand level per episode. This section asks a sharper, QoE-motivated question offline: under genuinely congested, dynamic, randomized multi-slice traffic with real shared-PRB contention (Eq. 3), does DQN preserve URLLC’s SLA compliance ahead of eMBB/mMTC, consistent with URLLC already carrying the highest reward priority (Table I)? This required two fixes to the base offline simulator: matching its ceiling headroom to the live config (Section IV-C’s finding), and adding the shared-pool constraint of Eq. 3 (calibrated to $B=8$, smaller than the three ceilings’ combined maximum of 19 PRB, so

TABLE III: SLA compliance (%) under congested/dynamic/randomized traffic with genuine shared-PRB contention; held-out seeds $\times 15$ episodes/arm, frozen weights. U = priority-weighted SLA utility, Eq. 4 (higher is better; baseline leads).

Arm	URLLC	eMBB	mMTC	U (%)
baseline (static)	22.6	34.5	19.0	27.2
DQN (SLA reward)	30.9	7.9	9.2	21.0
DQN (QoE reward)	27.0	8.2	10.8	19.0

maxing every ceiling simultaneously creates genuine scarcity), since the base simulator computed each slice’s served PRBs independently and never let one slice’s ceiling take capacity from another. Per-episode demand is additionally randomized ($0.4 \times -1.3 \times$ each slice’s mean) and remains dynamic within-episode.

Table III and Fig. 5 show the held-out result, trained with a corrected exploration schedule: an earlier version of this experiment left `DQNAdmissionPolicy`’s per-episode epsilon decay uncalled, so training-time action selection was uniform-random throughout (confirmed via checkpoint inspection, $\epsilon = 1.0$ after $\sim 18,000$ gradient steps) rather than the intended decay from full exploration to near-greedy; Q-learning is off-policy and still learns from the randomly-collected, reward-labeled transitions, but this is not the schedule Table I intends, so we fixed it and retrained both arms before reporting final numbers.¹ With the fix, both DQN arms improve URLLC above baseline (22.6% $\rightarrow$ 30.9% SLA-only, $\rightarrow$ 27.0% QoE-aware) by actively *sacrificing* eMBB (34.5% $\rightarrow$ 7.9%/8.2%) and mMTC (19.0% $\rightarrow$ 9.2%/10.8%) – a genuine, learned priority tradeoff, not a policy that merely rides every ceiling upward. This is no longer reward-mode-specific: an earlier, since-corrected run had shown only the QoE-shaped reward producing this tradeoff; with exploration correctly decaying, DQN (SLA) shows the same tradeoff as DQN (QoE), if anything a marginally larger URLLC gain – evidence the tradeoff is a property of a properly-trained DQN policy under this priority structure (Table I), not a QoE-reward-specific effect, a more robust but less differentiated finding than we initially reported. Measured against the priority-weighted utility U (Eq. 4), *baseline still leads* ($U=27.2\%$ vs. 21.0%/19.0%, Table III): eMBB’s declared priority weight (3.5) is large enough that the compliance it gives up outweighs URLLC’s gain under Table I’s actual weights. The tradeoff is real and reproducible, but not yet a utility improvement against this paper’s own declared objective – whether to revisit the tradeoff or the priority weights that make it utility-negative is open for the journal-length follow-up. This offline, preliminary result is not yet live-validated (Section VI).

¹A2C [15], also trained under this protocol (ϵ -schedule bug not applicable), converged to a fully deterministic, state-independent policy regardless of exploration/entropy interventions – an algorithm-dependent limitation, omitted from the headline comparison.

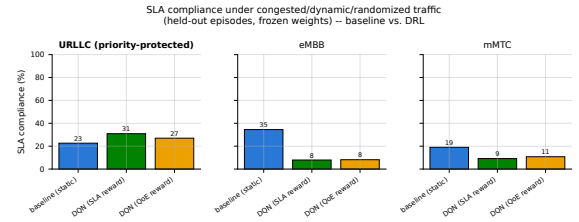


Fig. 5: SLA compliance under congested/dynamic/randomized traffic with genuine shared-PRB contention, baseline vs. DQN under both reward modes, URLLC panel bolded. Both DQN arms lead on URLLC and visibly sacrifice eMBB/mMTC to do it – not a QoE-reward-specific effect.

TABLE IV: This paper against [3]’s five gaps.

Gap	Status here
Validated deployment	Narrowed for slice admission control: real E2 loop, real MAC scheduler, real protocol stack; radio channel still simulated (rfsim) (Sec. IV-A–IV-D).
QoE inference	Partial : two calibration bugs found and fixed, corrected MOS no longer floor-pinned, but not yet re-validated at full statistical power live (Sec. IV-D).
Topology / scalability	Partial : single gNB; Sec. IV-E is a first offline contention result.
Privacy (FL)	Untouched : no federated learning; single-rig.
Resilience / scarcity	Partial : Sec. IV-E’s scarcity is offline-only; genuine live demand-driven contention was assessed and found infeasible on this rig’s current resource envelope (no spare per-slice UE provisioning, tight memory headroom) – documented, not silently skipped.

V. DISCUSSION

A. What This Paper Achieves Relative to Our Review

Our review [3] identified five recurring gaps – validated deployment, QoE inference, topology awareness, privacy preservation, and scalability – and proposed a QoE-aware GAT-MARL-FL framework conditioned on five validation directions. This paper **narrows** validated deployment for slice admission control: protocol stack, MAC scheduler, E2 loop, and core network real; radio channel still rfsim-simulated. Sections IV-A–IV-C show DRL-based admission, realized as ceiling nudging on real E2 hardware, reliably beats both a static ratio and a static-at-cap oracle under constant load, with the deployment engineering numbers already measured (Section IV-B). It contributes **partial** evidence toward QoE inference (Section IV-D found and fixed two calibration bugs; corrected MOS is no longer floor-pinned, but a fully-powered live re-run remains outstanding) and toward topology/scalability (Section IV-E is a first offline demonstration that shared-resource contention is learnable by a single-agent controller, not yet across multiple gNBs or with a topology-aware state). Privacy preservation (FL) remains **untouched**: single-rig, single-tenant by construction. Table IV states this plainly.

B. What's Next: Toward the Journal-Length Follow-Up

(1) Run the fully-powered live re-run under Section IV-D's correction (matching Section IV-A's protocol, currently $n=6-21/\text{arm}$) to settle whether static-at-cap's collapse-avoidance finding reaches significance under it. (2) Live-validate Section IV-E's shared-pool-contention finding, which needs new per-slice UE provisioning (IMSI/PDU-session, not a config change) and a memory-headroom fix on this rig first. (3) Replace the single-agent-per-slice arms with the GAT-encoded, centralized-training-decentralized-execution multi-agent controller our review proposes [5], motivated by Section IV-E's finding that a single-agent policy learns the URLLC-priority tradeoff only under careful, manual reward shaping; extend to multiple gNBs before layering on federated learning [6]. None of these three is attempted here; each is scoped by what Sections IV-A–IV-E establish, and is the direct target of our #5 journal-length follow-up.

VI. LIMITATIONS

Results are from OAI's rfsim radio simulator, not over-the-air deployment. The testbed runs on a single shared desktop machine, subject to intermittent radio-link failure under memory pressure, mitigated by automatic health-check-and-restart (verified to recover every time); every evaluation episode counted here completed and was log-verified end to end. A single gNB and three UEs were used, so paper #2's multi-gNB load-balancing term is not evaluated here. Live evaluation used 3 seeds per arm. Per-slice MOS is an inferred quantity, not a measurement from real end-user subjective scoring, and Section IV-A's headline table omits it rather than report the pre-correction value (Section IV-D). Section IV-D's corrected-calibration live numbers are directional only ($n=6-21/\text{arm}$, smaller than Section IV-A's fully-powered $n=15/\text{arm}$), so the static-at-cap collapse-avoidance finding is direction-replicated but not re-confirmed under the correction ($p=0.495$ vs. $p=0.0149$), not settled twice. Genuine live demand-driven scarcity for Section IV-E was assessed, not attempted, for the resource reasons in Table IV. Section IV-E's result is offline-only, evaluates DQN only (A2C's contrasting, degenerate behavior is reported in a footnote there but not part of this paper's live-validated results), and has not been reproduced on the live rig.

VII. CONCLUSION

This paper realizes the slice admission-control MDP from [1], [2] on a real OAI gNB over its native RIC-free E2 loop – protocol stack, MAC scheduler, core network real, radio channel still rfsim-simulated – narrowing our own systematic review's [3] validated-deployment gap, and centers evaluation on QoE rather than SLA compliance alone. Across 45 live episodes on real hardware, DQN holds $100.0 \pm 0.0\%$ SLA compliance under both SLA-only and QoE-aware rewards against a static baseline's seed-dependent $73.6 \pm 18.6\%$ (worst episode 0.0%); a static-at-cap oracle that removes the admission decision itself collapses at baseline's own rate (4/15 episodes, statistically tied), while DQN never collapsed once

across 25 reverified episodes ($p=0.0149$) – the mechanism is the admission decision, not the ceiling value. We disclose the engineering numbers a deployment needs (sub-ms E2 round trip, sub-100- μs inference, 5 s cadence held to within 1 ms) and fix two MOS-calibration bugs found mid-study; a directional live re-run under the correction reproduces this picture but does not yet re-confirm the oracle-collapse finding at full power. Comparing offline-simulated and live results side by side shows transfer holds exactly for eMBB but not for URLLC/mMTC, a diagnosed, fixable limitation of the offline training environment rather than a robustness result. A preliminary, offline-only follow-up under genuine cross-slice PRB contention shows a properly-trained DQN policy learns to preserve URLLC SLA compliance under either reward ($22.6\% \rightarrow 30.9\%$ SLA-only, $\rightarrow 27.0\%$ QoE-aware) by actively trading off eMBB ($\rightarrow 7.9\%/8.2\%$) and mMTC ($\rightarrow 9.2\%/10.8\%$) – the same URLLC-first objective as our prior simulated work [1], [2], now shown under real contention and, once a training-time exploration-schedule bug we fixed was corrected, a property of the trained policy rather than of the QoE-shaped reward, though not yet a win on this paper's own priority-weighted utility. The proposed GAT-MARL-FL framework [3] remains the journal-length follow-up's target, motivated by this paper's finding that single-agent control needs careful, manual reward shaping to achieve the URLLC-priority tradeoff.

REFERENCES

- [1] M. P. Kunasegran, W. L. Pang, and S. K. Phang, "Optimising resource allocation in 5G networks: Balancing URLLC and eMBB traffic under gNB congestion," in *2025 Multimedia University Engineering Conference (MECON)*, 2025, pp. 1–6.
- [2] —, "SLA-aware DRL for joint slice admission control and load balancing in multi-service 5G RAN," in *2025 9th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2025, pp. 207–212.
- [3] —, "Comprehensive review of optimisation techniques for user-centric distributed network slicing in 5G networks," *IEEE Access*, 2026, vVERIFY: final volume/issue/DOI once formally published – title page seen locally still shows placeholder publication-date/DOI fields.
- [4] X. Foukas, G. Patounas, A. Elmokashfi, and M. K. Marina, "Network slicing in 5G: Survey and challenges," *IEEE Communications Magazine*, vol. 55, no. 5, pp. 94–100, 2017.
- [5] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.
- [6] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [7] WinesLab, "ORANSlice: O-RAN network slicing on OpenAirInterface," <https://github.com/wineslab/ORANSlice>, oRANSlice fork, OAI 2024.w28 base, commit b9bcc9b. VERIFY: confirm citation format WinesLab/Northeastern prefers for this repo.
- [8] OpenAirInterface Software Alliance, "OpenAirInterface: 5g software alliance for democratising wireless innovation," <https://openairinterface.org>, vVERIFY: check for a canonical OAI citation (e.g. a EuCNC/WCNC demo paper) vs. the project URL.
- [9] O-RAN Alliance, "O-RAN: Towards an open and smart RAN," White Paper, 2018.
- [10] —, "O-RAN working group 3, near-RT RIC architecture and E2 general aspects and principles (E2AP)," O-RAN Alliance, Tech. Rep., 2023, vVERIFY: exact spec number/version (e.g. O-RAN.WG3.E2AP-v03.00) and year.

- [11] —, “O-RAN working group 3, E2 service model (E2SM) KPM,” O-RAN Alliance, Tech. Rep., 2023, vERIFY: exact spec number/version (e.g. O-RAN.WG3.E2SM-KPM-v03.00) and year.
- [12] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski *et al.*, “Human-level control through deep reinforcement learning,” *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [13] M. Fiedler, T. Hossfeld, and P. Tran-Gia, “The IQX hypothesis: A fundamental property of quality of experience,” *IEEE Network*, 2010, vERIFY: exact volume/issue/page numbers.
- [14] ITU-T, “P.1203: Parametric bitstream-based quality assessment of progressive download and adaptive audiovisual streaming services over reliable transport,” ITU-T, Tech. Rep., 2017, vERIFY: exact edition/amendment referenced (base P.1203 vs. P.1203.1/.2/.3).
- [15] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, “Asynchronous methods for deep reinforcement learning,” in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016, pp. 1928–1937.